



Semester 1 Examinations 2013 - 2014

Exam Code	1SD1, 1MDM1, 1IF1, 1MF1, 1MCR1, 1SD3
Exam	Higher Diploma in Software Design and Development Higher Diploma in Software Design and Development, Industry Stream Masters in Information Technology Masters in Software Design and Development Masters in Digital Media Masters Clinical Research
Module Code	CT511
Module	Database Development
External Examiner	Professor Liam Maguire
Internal Examiners	*Ms. Pat Byrne Professor G. Lyons Dr. Michael Madden
<u>Instructions</u>	Answer question 1 and any two others
Duration	2 hours
No. of Pages	3
Department	Information Technology
Requirements	None

1. (a) A small radio station plans to create a website which lists details on all of the programmes they transmit, including the date and time, the presenter's name, topics discussed and music tracks played. This information is displayed on their website and can be accessed by searching on the time of the show or the presenter. Information on each track played also needs to be held for the Irish Musical Right organisation (IMRO) who are responsible for paying royalties to the copyright owners of any music played.

The radio station requires a database to manage all of this data. Draw up the appropriate normalised tables to hold all the required data, indicating primary and foreign keys in each case. Document any assumptions you make in your design. [20]

- (b) A company uses a database to process travel expenses incurred by its employees. The following tables are included in this database (primary keys are indicated in bold):

EMPLOYEE	Emp_ID	Emp_name	Dept_code	
DEPARTMENT	Dept_code	Dept_name	Dept_manager	
CLAIM	Claim_num	Emp_ID	Claim_date	Claim_paid
EXPENSE	Claim_num	Exp_num	Exp_type	Exp_amount

Note: Each claim must be signed by the departmental manager before submission.
 A claim may have a number of expense items for different costs incurred.
 Claim_paid indicator contains a value of 'U' until the money has been transferred to the employee's account, when it is changed to 'P'.
 Exp_type indicates the type of expense claimed, and may have a value of 'T' for travel, 'A' for accommodation, 'M' for meals and 'O' for other.

Create SQL queries to satisfy the following requests:

- List the name and department of all employees who have made a claim.
- List details of all outstanding claims for employees of the Accounts department.
- List the ID of employees who have incurred 'other' expense which exceed €20.
- Total the expense items not yet paid, with sub totals by department.
- List all claims for manager Fergal Anderson's department during October 2013.

[20]

2. (a) Define the rules of normalisation to third normal form, in each case explaining how the rules are used to prevent anomalies in the data. Use an example to illustrate your answer [15]
- (b) Explain how the problems of lost updates, uncommitted data and inconsistent retrieval can occur in a database transaction. How might these problems be overcome? [15]
3. (a) Explain the following terms when used with databases:
tuple ; foreign key ; metadata ; denormalisation ; referential integrity [20]
- (b) Why employ fragmentation in a distributed database? Explain some of the techniques by which this might be carried out successfully. [10]
4. (a) You have been asked by a small local company which sells oriental carpets to help them in setting up a database to monitor their sales. Explain the steps you might take in your work with them. [20]
- (b) Distinguish between the types of data held in a data warehouse and a conventional transactional database, and describe the steps which should be taken in setting up a data warehouse. [10]